

YAOSHENG XU

PERSONAL INFORMATION

Email: Yaosheng.Xu@chicagobooth.edu
Phone: +1 (607) 280-2760
Current position: Principal Researcher *Advisor: Barış Ata*
 Booth School of Business
 University of Chicago

EDUCATION

Cornell University Ph.D. in Statistics. <i>Advisor: Jim Dai</i>	08/2017 – 08/2022
Shandong University M.S. in Probability and Mathematical Statistics	09/2014 – 07/2017
Shandong University B.S. in Statistics	09/2010 – 08/2014

WORK EXPERIENCE

Principal Researcher. Full-time, Booth School of Business, University of Chicago.	06/2024 – present
Applied Scientist. Full-time, Amazon.com.	08/2022 – 05/2024
Applied Scientist Intern. Amazon.com.	06/2021 – 08/2021

PUBLICATIONS AND PREPRINTS

- Barış Ata and Yaosheng Xu.
Dynamic Control of a Stochastic Matching System in Heavy Traffic: An Effective Computational Method for High-Dimensional Problems. Under review at **Operations Research**. [Link](#)
- Yaosheng Xu.
Explicit Steady-State Approximations for Parallel Server Systems with Heterogeneous Servers. Under review at **Operations Research**. [Link](#)
- J. G. Dai, Peter Glynn and Yaosheng Xu.
Asymptotic Product-Form Steady-State for Generalized Jackson Networks in Multi-Scale Heavy Traffic. Accepted at **Operations Research**. [Link](#)
- Jin Guang, Yaosheng Xu, and J. G. Dai.
Steady-State Convergence of the Continuous-Time Routing System with General Distributions in Heavy Traffic.
ACM SIGMETRICS, 9(1): 1-29, 2025. [Link](#)

A short version is accepted at Workshop on Mathematical Performance Modeling and Analysis (MAMA) of ACM SIGMETRICS 2024. [Link](#)
- Xin Bing, Yang Ning and Yaosheng Xu.
Adaptive Estimation in Multivariate Response Regression with Hidden Variables.
Annals of Statistics, 50(2): 640–672, 2022. [Link](#)

6. Mondher Bellalah, Yaosheng Xu, Detao Zhang.
Intertemporal Optimal Portfolio Choice based on Labor Income within Shadow Costs of Incomplete Information and Short Sales.
Annals of Operations Research, 281: 397–422, 2019. [Link](#)

WORK IN PROGRESS

1. Barış Ata and Yaosheng Xu.
Dynamic Dispatch and Relocation Control for Ride-Hailing Systems Incorporating Travel Times.

WORKSHOPS AND CONFERENCE PRESENTATIONS

1. INFORMS APS Conference. Invited Talk. Atlanta, GA. June 30 – July 3, 2025.
Title: Dynamic Control of Stochastic Matching Systems in Heavy Traffic: An Effective Computational Method for High-Dimensional Problems
2. INFORMS Annual Meeting. Invited Talk. Seattle, WA. October 20 – 23, 2024.
Title: Explicit Steady-State Approximations for Parallel Server Systems with Heterogeneous Servers and Class-dependent Service Time
3. INFORMS APS showcase. Flash Talk. Seattle, WA. October 22, 2024.
Title: Asymptotic steady-state independence for generalized Jackson Networks in multi-scale heavy traffic
4. INFORMS Annual Meeting. Invited Talk. Anaheim, CA. October 24 – 27, 2021.
Title: Explicit Steady-state Approximations For Parallel-server Systems
5. Fields-China Joint Industrial Problem Solving Workshop in Finance. Toronto, ON. May 8 – 12, 2017.
Title: Influence Factors for the Chinese Stock Index Futures Basis

RESEARCH INTERESTS AND SKILLS

Operations Research and Management

Stochastic Optimization and Control, Dynamic Matching, AI-driven Diffusion Control, Stochastic Processing Networks, Queueing Theory, Multi-Scale Heavy Traffic, Load-balancing Systems, LLM Inference.

Statistics and Data Science

High-Dimensional Statistics, Multivariate Analysis, Causal Inference, Model Selection and Regularization, Experimental Design.

Programming Languages

Python, C++, R, MATLAB.

TEACHING EXPERIENCE

Supervised courses as Teaching Assistant at Cornell University

- STSCI/BTRY 3080: Probability Models and Inference
- ORIE 3510/5510: Introduction to Engineering Stochastic Processes
- STSCI 4740: Data Mining and Machine Learning
- BTRY 6020: Statistical Methods
- STSCI 2150: Introductory Statistics for Biology

REFERENCES

Prof. Jim Dai

Leon C. Welch Professor of Engineering
Operations Research and Information Engineering
(ORIE)
Cornell University
E-mail: jd694@cornell.edu
Phone: +1 (607) 379-1922

Prof. Peter Glynn

Thomas W. Ford Professor
Department of Management Science & Engineering
Stanford University
E-mail: glynn@stanford.edu
Phone: +1 (650) 725-1624

Prof. Baris Ata

Sigmund E. Edelstone Distinguished Service Profes-
sor
Booth School of Business
University of Chicago
E-mail: Baris.Ata@chicagobooth.edu
Phone: +1 (773) 807-9107

Prof. Yang Ning

Associate Professor
Department of Statistics and Data Science
Cornell University
E-mail: yn265@cornell.edu
Phone: +1 (607) 255-3759